

BORDEAUX MERIGNAC, FranceWMO: **075100**Lat: **44.8308N**Lon: **0.6914W**Elev: **162**StdP: **14.61**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.2	29.0	19.5	14.8	32.2	22.9	17.3	33.9	27.6	54.1	24.0	53.2	5.0	20	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	91.6	70.5	87.9	68.8	84.4	67.4	72.7	86.6	71.0	83.6	69.4	80.3	8.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	105.1	77.1	66.9	99.6	75.4	65.6	95.0	74.0	36.3	86.8	34.9	84.1	33.5	80.3	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.6	16.2	22.1	97.8	3.5	2.8	19.6	99.8	17.5	101.5	15.6	103.1	13.1	105.1		
(5)			21.1	75.8	3.6	1.7	18.6	77.0	16.5	78.0	14.5	78.9	11.9	80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	44.5	45.6	50.8	54.8	61.1	67.2	70.4	70.8	65.2	59.3	50.0	45.3	(6)
(7)		DBStd	11.34	6.80	6.58	5.86	5.76	6.26	6.39	5.80	5.70	5.55	6.24	6.28	7.19	(7)
(8)		HDD50	677	190	147	62	17	2	0	0	0	0	8	76	175	(8)
(9)		HDD65	3475	635	544	440	309	155	48	16	11	64	192	451	610	(9)
(10)		CDD50	3289	20	23	87	162	346	516	632	644	456	297	75	31	(10)
(11)		CDD65	611	0	0	0	4	35	113	183	190	70	16	0	0	(11)
(12)		CDH74	5492	0	0	5	65	352	1095	1669	1626	593	86	1	0	(12)
(13)		CDH80	1945	0	0	0	8	85	399	635	646	162	10	0	0	(13)
(14)	Wind	WSAvg	7.6	7.9	8.3	8.6	8.4	7.8	7.4	7.1	6.7	6.6	6.9	7.4	7.6	(14)
(15)	Precipitation	PrecAvg	36.2	3.7	3.0	2.7	2.8	2.9	2.4	1.9	2.2	3.1	3.4	4.1	3.8	(15)
(16)		PrecMax	48.1	9.2	6.1	7.5	8.0	6.2	6.6	4.4	9.8	9.8	8.7	12.7	12.0	(16)
(17)		PrecMin	23.1	0.4	0.1	0.0	0.1	0.3	0.1	0.2	0.2	0.1	0.2	0.4	0.1	(17)
(18)		PrecStd	6.9	2.2	1.5	1.6	1.8	1.5	1.5	1.0	1.8	2.2	2.2	2.6	2.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	66.2	74.2	80.8	87.6	95.1	95.3	97.1	89.5	81.2	69.1	62.0	(19)
(20)			MCWB	54.2	54.1	57.0	61.8	66.4	72.6	71.0	72.1	67.8	65.2	60.3	54.7	(20)
(21)		2%	DB	58.0	61.2	69.4	74.9	82.0	88.6	90.6	91.8	84.5	76.2	64.4	58.9	(21)
(22)			MCWB	53.3	52.1	55.2	57.9	64.1	68.9	70.1	71.3	66.5	64.8	58.4	54.4	(22)
(23)		5%	DB	55.9	57.8	65.2	70.8	77.8	84.5	86.9	87.1	80.7	72.3	61.4	56.8	(23)
(24)			MCWB	52.3	51.1	53.6	56.9	62.8	67.7	68.8	69.1	65.5	62.8	56.8	53.5	(24)
(25)		10%	DB	53.9	55.1	61.4	66.4	73.6	80.4	83.0	82.7	77.0	69.1	59.2	55.2	(25)
(26)			MCWB	51.0	50.5	51.9	55.1	60.9	65.9	67.3	67.6	64.1	61.5	55.6	52.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.2	55.8	58.9	63.1	68.9	74.0	74.4	75.7	71.4	68.0	62.1	57.2	(27)
(28)			MCD B	58.4	61.4	69.9	76.5	82.4	88.7	89.8	89.8	83.2	74.9	66.1	59.0	(28)
(29)		2%	WB	54.6	54.0	56.6	60.4	66.3	71.3	72.1	72.9	69.2	65.9	59.4	55.6	(29)
(30)			MCD B	57.0	58.4	65.7	71.3	79.0	85.3	86.4	86.1	79.3	73.0	62.5	57.6	(30)
(31)		5%	WB	53.0	52.7	55.0	58.1	64.0	69.2	70.3	71.1	67.2	64.3	57.8	54.2	(31)
(32)			MCD B	55.3	56.5	62.3	67.6	74.4	80.9	82.7	82.9	76.4	70.0	60.8	56.4	(32)
(33)		10%	WB	51.3	51.2	53.4	56.0	62.0	67.1	68.7	69.2	65.5	62.6	56.0	52.6	(33)
(34)			MCD B	53.7	54.6	59.2	64.0	71.2	77.4	79.8	79.2	74.1	68.2	59.0	55.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.0	14.7	17.3	18.0	18.4	19.5	19.7	20.1	19.7	16.6	13.5	12.1	(35)
(36)			MCDBR	13.2	18.9	24.5	25.4	25.0	27.0	26.4	27.0	25.6	20.3	15.3	12.6	(36)
(37)			MCWBR	9.3	11.3	12.4	11.8	10.9	10.8	9.9	9.7	10.4	10.3	9.7	8.8	(37)
(38)		5% WB	MCDBR	11.3	16.0	19.9	21.8	22.4	23.2	23.4	23.0	21.4	17.5	13.4	11.5	(38)
(39)	MCWBR		8.9	10.8	10.9	10.9	10.6	10.4	9.9	9.4	9.9	9.6	9.3	8.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.341	0.373	0.394	0.400	0.413	0.403	0.400	0.373	0.368	0.343	0.325	(40)	
(41)		taud	2.481	2.429	2.345	2.292	2.329	2.325	2.348	2.371	2.428	2.448	2.463	2.477	(41)	
(42)		Ebn at Noon	250	267	272	275	276	272	273	270	269	253	240	238	(42)	
(43)		Edh at Noon	22	28	35	39	39	40	38	36	31	27	23	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	417	725	1105	1488	1741	1954	1916	1707	1367	860	503	384	(44)	
(45)		RadStd	40	82	132	148	134	137	121	107	87	71	61	47	(45)	